

DRY-RUN GUIDE | leap\_year.java  
Source: java\1st\_belt\leap\_year.java

#### CORE BEHAVIOR

Reads an integer if available. A year is leap when divisible by 4 and not 100, or divisible by 400.

#### REPRESENTATIVE INPUT / CALL

Input: '1900'.

#### STEP TRACE

1900%4=0 but 1900%100=0, so the first conjunction is false; 1900%400 is not zero.

#### EXPECTED OUTPUT / RETURN

'not leap year'

#### ACTUAL SOURCE EXCERPT

Verbatim lines from the source; long lines wrap. Source files were not modified.

```
001 | /*
002 | Challenge #3: Leap Year
003 |
004 | You are given a year as input. A leap year is defined as a year that has 366 days.
005 | Write a program that outputs "leap year" if the input year is a leap year and "not
    | leap year" otherwise.
006 |
007 | A year is a leap year if it satisfies the following conditions:
008 | - It is divisible by 4.
009 | - If it is divisible by 100, it must also be divisible by 400.
010 |
011 | Example 1:
012 | Input: 2020
013 | Output: leap year
014 | */
015 |
016 | import java.io.*;
017 | import java.util.*;
018 |
019 | public class Solution {
020 |     public static void main(String[] args) {
021 |         Scanner sc = new Scanner(System.in);
022 |         if (sc.hasNextInt()) {
023 |             int year = sc.nextInt();
024 |
025 |             // A year is a leap year if it is divisible by 4,
026 |             // but not by 100 unless it is also divisible by 400.
027 |             if ((year % 4 == 0 && year % 100 != 0) || (year % 400 == 0)) {
028 |                 System.out.println("leap year");
029 |             } else {
030 |                 System.out.println("not leap year");
031 |             }
032 |         }
033 |         sc.close();
034 |     }
035 | }
036 |
```